

Distributed EV Charging Navigation System

FIT3143 Applied #1 - Network topology, MPI architecture, and communication analysis

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Executive decision. We model three different communication scopes: a non-periodic 2-D logical mesh for local charging-node sensing, a region-sharded star forest for direct node-to-base reporting, and a small base-station control overlay for global redirection. The mesh is a logical regular-area baseline, not a claim that physical EV stations form a grid.

1. Task 1 - Network topology

The 3 x 3 drawing in the specification is illustrative. Physical station placement is external to this assessment; the design task is to choose logical communication relationships. Immediate adjacency is logical: it neither guarantees the geographically closest station nor asserts a dedicated physical cable. The simulator must distinguish physical deployment, logical topology, and MPI process placement.

Topology	Degree	Diameter	Cost	EVCNS fit
Linear array	1-2	N-1	N-1	Poor 2-D locality; O(N) remote path.
Ring	2	$\text{floor}(N/2)$	N	Only two neighbours; artificial wrap-around.
Star	1 / N-1	2	N	Good reporting hub, not local sensing.
Binary tree	1-3	$O(\log N)$	N-1	Parent-child is not geographic locality.
2-D mesh	2-4	$O(\sqrt{N})$	O(N)	Selected local logical baseline.
2-D torus	4	$O(\sqrt{N})$	O(N)	False boundary neighbours.
3-D mesh / cube	3-6	$O(N^{1/3})$	O(N)	Unnatural third locality dimension.
Hypercube	$\log_2(N)$	$\log_2(N)$	$N \log_2(N)/2$	Non-geographic edges; growing degree.
Fully connected	N-1	1	$N(N-1)/2$	Not scalable for large N.

The selected hybrid topology applies each topology only where it fits: mesh for local sensing, regional stars for reporting, and a small inter-base control overlay for global coordination. For an $R \times C$ mesh, $E = R(C - 1) + C(R - 1) = 2N - R - C$, so the local link cost grows linearly. A strongly irregular deployment is an explicit future extension using a configured adjacency graph.

2. Task 2 - MPI simulation architecture

Each ChargingNode contains its own ChargingPort objects. A port is local shared-memory state rather than a separate MPI rank. A BaseStation owns a region, maintains a cache and event log, and coordinates redirects.

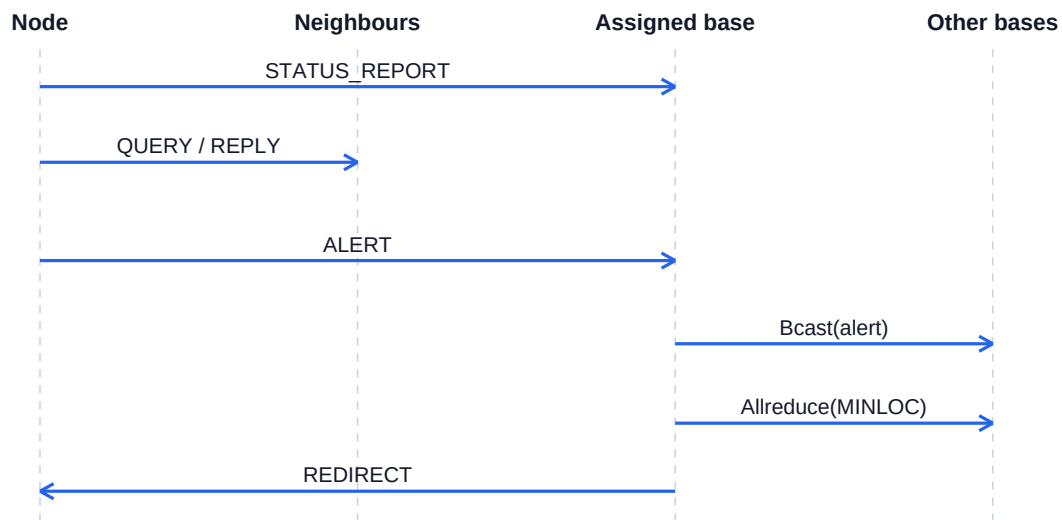


Message / operation	Direction	Trigger	Purpose
STATUS_REPORT	node -> assigned base	Every round	Refresh cache and log.
QUERY	heavy node -> local neighbours	utilisation > threshold	Request local utilisation.
NEIGHBOUR_REPLY	neighbour -> querying node	On QUERY	Return availability and utilisation.
ALERT	node -> assigned base	node and all valid neighbours heavy	Report local saturation.
Bcast + Allreduce(MINLOC)	bases	Every alert	Select globally nearest available candidate.
REDIRECT	owner base -> alerting node	Candidate selected	Return target or no availability.

Ranks 0 ... S-1 represent base stations and ranks S ... S+N-1 represent charging nodes. node_comm contains charging nodes; cart_comm is its non-periodic Cartesian topology; base_comm contains bases. MPI provides inter-process communication. OpenMP is used only when a sufficiently large ports[P] scan benefits from local shared-memory parallelism; MPI calls remain master-thread only under MPI_THREAD_FUNNELED.

3. Task 2 - communication sequence

Every node reports status each round. Neighbour queries occur only when the local utilisation is above the user-configured threshold. Missing or stale neighbour data never proves all neighbours are busy; it produces an incomplete assessment and withholds the all-neighbours alert.



For every alert, every base supplies its regional nearest candidate. This is required because an available candidate in the source region is not necessarily globally closest. Allreduce(MINLOC) therefore remains in the correctness path.

4. Task 3 - communication analysis

The cluster is modelled with 1 Gbps links. For a payload L bytes and a fabric hop count H_{fabric} that is constant with respect to N , the serialisation component is $T_{\text{tx}} = 8 L H_{\text{fabric}} / B$. In the figures, $H_{\text{fabric}} = 1$ is an equivalent-link assumption. Real MPI wall-clock time additionally includes startup, switching, contention, and queueing.

Message	Payload assumption	Scaling result
QUERY	16 B	One transfer $O(1)$; total local traffic $O(N)$.
NEIGHBOUR_REPLY	32 B	One transfer $O(1)$; total local traffic $O(N)$.
STATUS_REPORT	40 B	One transfer $O(1)$; per-base ingress $O(N/S)$.
ALERT	24 B	$O(A)$ traffic per round, where $0 \leq A \leq N$.
REDIRECT	32 B	$O(A)$ traffic per round.
Bcast + MINLOC	24 B + 16 B	$O(\log S)$ critical path per alert.

For a mesh with $E = 2N - R - C$, the worst-case local message count is $2E$ QUERYs plus $2E$ replies: $4E = O(N)$. If A alerts occur in one round, tree-based collective coordination has critical path $O(A \log S)$. In the worst case $A = N$, this is $O(N \log S)$. More bases reduce regional ingress from approximately N to N/S but increase collective depth logarithmically; the outcome is diminishing returns. When $S = 1$, coordination is a singleton operation and the only base cache is global.

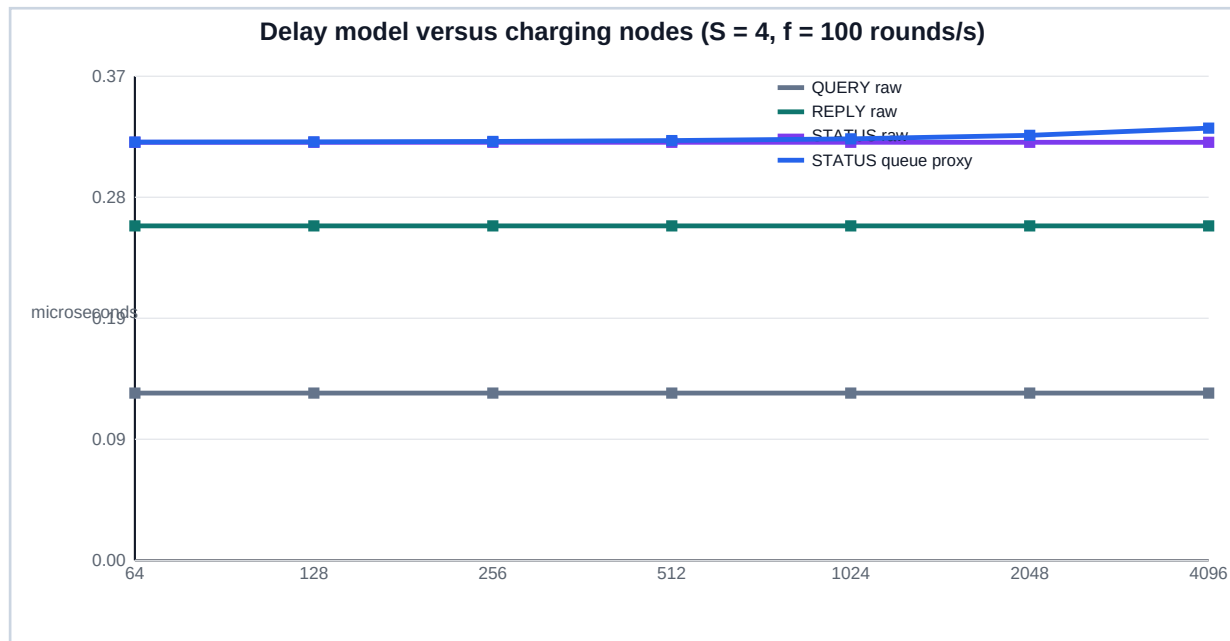


Figure 1. Raw serialisation is constant per direct message; the blue curve is a stated base-ingress queue proxy.

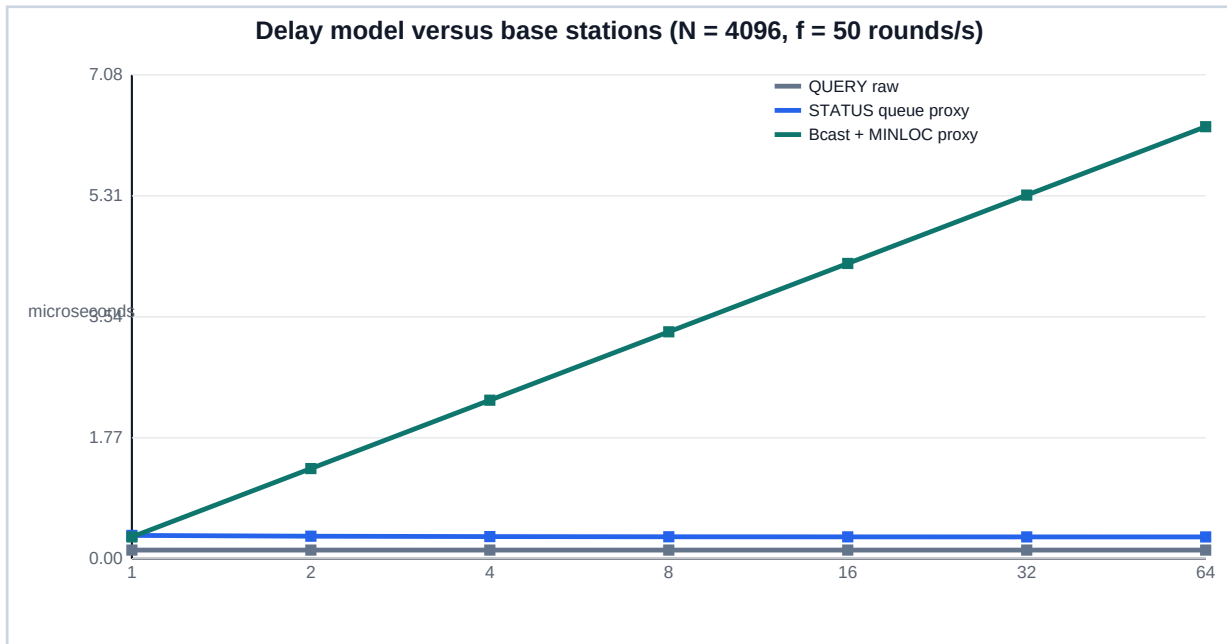


Figure 2. More bases reduce report ingress pressure, while collective coordination depth grows.

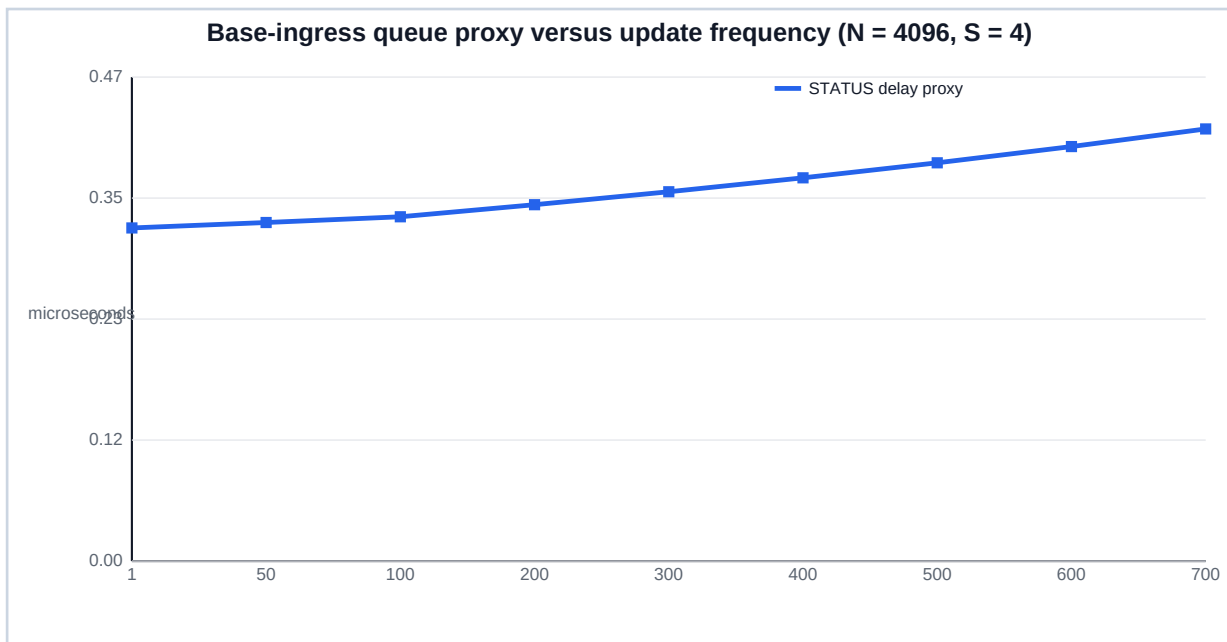


Figure 3. Higher update frequency increases offered base load and queueing near capacity.

5. Limitations, future work, and references

This report makes a logical regular-area mesh assumption and uses analytical payload/queueing models. It does not replace measurements on the target cluster. A production system would add irregular configured adjacency graphs, stale-data expiry, failure detection, authentication, and durable storage.

Presentation conclusion

We use a non-periodic 2-D logical mesh for bounded local sensing, regional stars for node reporting, and a small control overlay for globally correct redirection. This hybrid architecture models each communication scope with a topology appropriate to its scale and purpose.

References

- FIT3143 Applied #1 - Distributed Wireless Sensor Network assessment specification.
- FIT3143 Applied #1 marking rubric.
- FIT3143 Topic 5 - network topology and MPI material.
- Teaching-team Ed clarification: 3 x 3 is illustrative; immediate adjacency is required; physical EV nodes are not constructed by students; choose a topology or hybrid topology for a real-life situation.

Before submission: replace student placeholders and attach the required AI declaration and complete prompt records.